MIL-W-14742A (MU)
17 January 1972
Superseding
MIL-W-14742 (MU)
3 June 1968

MILITARY SPECIFICATION

WARHEAD SECTION, GUIDED MISSILE, PRACTICE, M220, ASSEMBLY AND PACKAGING

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packaging of one type of warhead designated as Warhead Section, Guided Missile, Practice, M220.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-T-106 - Tape, Pressure-Sensitive, Adhesive Masking, Paper
UU-T-450 - Tissue, Facial

MILITARY

MIL-C-45662 - Calibration System Requirements

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition

FSC: 1336

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

PICATINNY ARSENAL

9260040 - Warhead Section, GM, Practice, M220

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and Assemblies.-The components and assemblies shall comply with all requirements specified on drawing (dwg.) 9260040 and with all requirements specified in applicable specifications.

3.3 Warhead to Missile Assembly Capability.-The warhead shall exhibit capabilities of assembly to the missile when tested as specified in 4.4.1.

3.4 Adhesion and Mar Resistance of Paint.-The paint on the warhead shall not exhibit flaking, peeling, loss of adhesion, marring or discoloration when tested as specified in 4.4.3.

3.5 Workmanship.-All parts and assemblies shall be fabricated in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous. Light scratches not exposing base material shall be ignored. All required markings shall be neat and sharply defined.

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3.6 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Submission of Product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample to a Government approved facility as designated by the contracting officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of three (3) warhead sections and three (3) complete sets of parts which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and shall submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance for materials (see 4.1.1.c.). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be Performed.-Assemblies and components will be subjected by the Government to any or all of the examinations and tests specified in 4.3.2 and 4.3.3 of this specification and to any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample, or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

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4.3 Inspection Provisions

4.3.1 Lot Formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with MIL-STD-1168. Each inspection lot of complete warhead section metal parts assemblies shall contain:

a. Ogives from not more than one interfix lot number from one manufacturer.
b. Bodies from not more than one interfix lot number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Warhead Section (see dwg. 9260040).

Categories	Defects	Method of Inspection	Code No. (see 6.3.4)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Length from tip of ogive to missile mating shoulder, max.	Gage	01001
102.	Total runout of ogive to Datums A and B	Gage	01002
Minor:	AQL 1.50 percent		
201.	Marking missing or illegible	Visual	01003
202.	Protective coating not as specified	Visual	01004
203.	Evidence of poor workmanship (see 3.5)	Visual	01005

4.3.2.2 Body (see dwg. 9260039).

Categories	Defects	Method of Inspection	Code No. (see 6.3.4)
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Circular runout of ogive mating diameter to Datum B	Gage	02001
102.	Circular runout of missile mating diameter to Datum A	Gage	02002
103.	Outside diameter	Gage	02003
104.	Length of missile mounting flange	Gage	02004
105.	Perpendicularity of ogive mating shoulder to Datum A	Gage	02005
106.	Perpendicularity of missile mating shoulder to Datum A	Gage	02006
107.	Presence of chromate finish	Visual	02007
108.	Inside corner radius of missile skin mating shoulder, max.	Gage	02008

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	02009

4.3.2.3 Ogive, Missile (see dwg. 9260038).

Categories	Defects	Method of Inspection	Code No. (see 6.3.4)
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Total runout of ogive nose to Datums A and B	Gage	03001
102.	Circular runout of ogive outside diameter to Datum A	Gage	03002
103.	Length of ogive	Gage	03003
104.	Contour of ogive	Gage	03004
105.	Perpendicularity of ogive base to Datum A	Gage	03005
106.	Presence of protective finish	Visual	03006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	03007

4.3.3 Testing

4.3.3.1 First Article Sample (see 4.2.1).-The units and parts submitted as the first article sample shall be tested as specified below after having been examined for requirements as specified in 4.2.2. Rejection and resubmission shall be in accordance with 4.2.3.

4.3.3.1.1 Weight (see dwg. 9260040 and 4.3.3.2.1).

4.3.3.1.2 Radial Center of Gravity (see dwg. 9260040)-Minor Defect-Code No. 01009.-This test shall be performed on each assembly using equipment in accordance with 4.3.4.

4.3.3.1.3 Longitudinal Center of Gravity (see dwg. 9260040) -Minor Defect-Code No. 01010.-This test shall be performed on each assembly using equipment in accordance with 4.3.4.

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4.3.3.1.4 Functional Examination for Missile Assembly (see 3.3 and 4.3.3.2.4).

4.3.3.1.5 Ogive-Body Joint Torque Test (see dwg. 9260040 and 4.3.3.2.5).

4.3.3.1.6 Adhesion and Mar Resistance of Paint (see 3.4) -Major Defects-Code Nos. 01006 and 01007.-These tests shall be performed on each assembly. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.2 Production

4.3.3.2.1 Weight (see dwg. 9260040) -Major Defect-Code No. 01008.-This determination shall be performed 100 percent using test equipment in accordance with 4.3.4. Any assembly failing to meet the requirement shall be classed defective and removed from the lot.

4.3.3.2.2 Functional Examination for Missile Assembly (see 3.3) -Major Defect-Code No. 010 11.-This test shall be performed 100 percent as specified in 4.4.1 using equipment in accordance with 4.3.4. Any assembly not conforming to the requirements shall be classed defective and removed from the lot.

4.3.3.2.3 Ogive-Body Joint Torque Test (see dwg. 9260040) -Major Defect-Code No. 01012.-This test shall be performed 100 percent as specified in 4.4.2 using equipment in accordance with 4.3.4. Any assembly not conforming to the requirements shall be classed defective and removed from the lot.

4.3.3.2.4 Adhesion and Mar Resistance of Paint (see 3.4) -Major Defects-Code Nos. 01006 and 01007.-The sampling plan for these tests shall be in accordance with MIL-STD-105, using Inspection Level II with an AQL of 0.40 percent. An inspection lot for this examination shall consist of those warheads or components painted in one day. These tests shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

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4.3.4 Inspection Equipment.-The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see 6.3).

4.3.4.1 Inspection Equipment Capability and Accuracy.-All inspection equipment and facilities shall be of adequate quality and quantity to perform properly and accurately all the required inspections, measurements and tests. The inspection equipment shall be maintained in satisfactory condition for the continued proper performance of these functions and shall be calibrated in accordance with the provisions of MIL-C-45662.

4.3.4.2 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements herein may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test Methods and Procedures

4.4.1 Functional Examination for Missile Assembly.-The warhead body shall be mounted in the test fixture and the following attribute examinations shall be performed: location of missile mating holes, diameter of missile mating flange, location of countersink (one hole only), pitch diameter and minor diameter of mating hole threads.

4.4.2 Ogive - Body Joint Torque Test.-The warhead section shall be mounted in the test fixture that will restrain the body from turning, and the ogive shall be subject to the minimum specified torque. Observation of ogive movement shall be noted.

4.4.3 Adhesion and Mar Resistance of Paint

4.4.3.1 Adhesion.-Firmly apply one (1) inch-wide strip of masking tape conforming to UU-T-106 for the full length of one (1) of the launch leader bonding areas and peel the tape off in one abrupt motion.

4.4.3.2 Mar Resistance.-Lay a small pad of tissue conforming to the requirements of UU-T-450 approximately one (1) inch square and 8 to 12 layers thick on one (1) of the launch leader bonding areas, other than the area subjected to the adhesion test.

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Fully saturate the pad with thinner utilized in the painting process. Cover the pad with a small beaker. Remove the pad after one (1) minute and rub the moistened spot with a clean dry cloth or tissue. Failure is indicated by discoloration of test cloth or tissue or by visible evidence of damage to the painted surface. A slight whitening of the surface which may be removed by light rubbing with a dry cloth shall not be reason for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking.-The parts shall be packaged, packed and marked in accordance with MIL-STD-1169.

6. NOTES

6.1 Intended Use.-The Practice Warhead Section, M220 covered by this specification is intended for use in the TOW Practice Guided Missile, BTM71.

6.2 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be perpared for each lot in accordance with the information specified in MIL-STD-1167.

c. Inspection equipment design requirements (see 4.3.4 and 6.3).

6.3 Inspection Equipment Designs

6.3.1 The contractor shall prepare designs or specify commercial equipment as required for all inspections described in paragraphs 4.3.2 and 4.3.3.

6.3.2 Contractor designs shall include, as applicable, operating instructions, calibration procedures and maintenance procedures.

6.3.3 Commercial equipment shall be fully described by means of catalogue sheets or the like. This information will include illustrations and engineering data adequate for evaluation. Designs shall be prepared of any special fixture required to be used with commercial equipment.

6.3.4 The specification number and the applicable five-digit defect code number from Section 4 of this specification shall be referenced on each contractor design.

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6.3.5 Designs shall be prepared to the category and form (per MIL-D-1000) specified in the Contract Data Requirements List (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance.

6.3.6 Submission for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval within the time frame and with the distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal reviews will be accomplished normally within one month after receipt.

6.4 Submission of First Article.-Instructions for the submission and location for testing and evaluation of the first article sample shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI, Dover, New Jersey 07801.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1336-A-168

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM, JAN 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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